

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DAA(01CT0512)	Aim: To implement NAND & NOR gate using CMOS transistors and simulate the design.	
Experiment:- 4	Date:- 06-08-2025	Enrollment No:- 92301733056

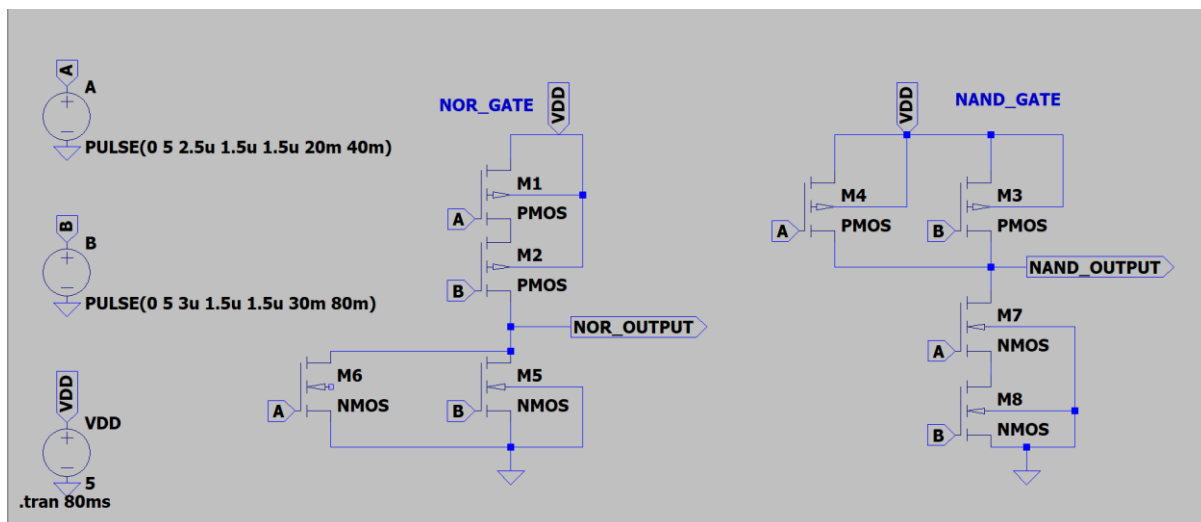
CO1: Develop fundamentals in fabrication of MOSFETs and MOS transistor

CO2: Analyze MOS inverters with its static and switching characteristics

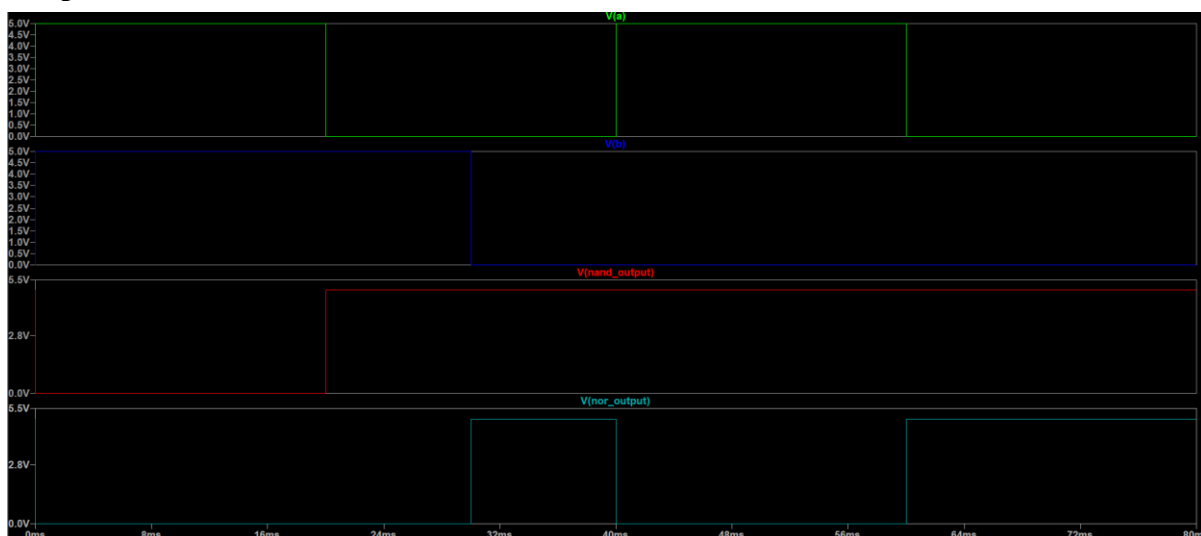
CO3: Analyze Combinational and Sequential MOS logic circuit

Aim: To implement NAND & NOR gate using CMOS transistors and simulate the design.

Circuit:



Output Waveform



Subject: DAA(01CT0512)

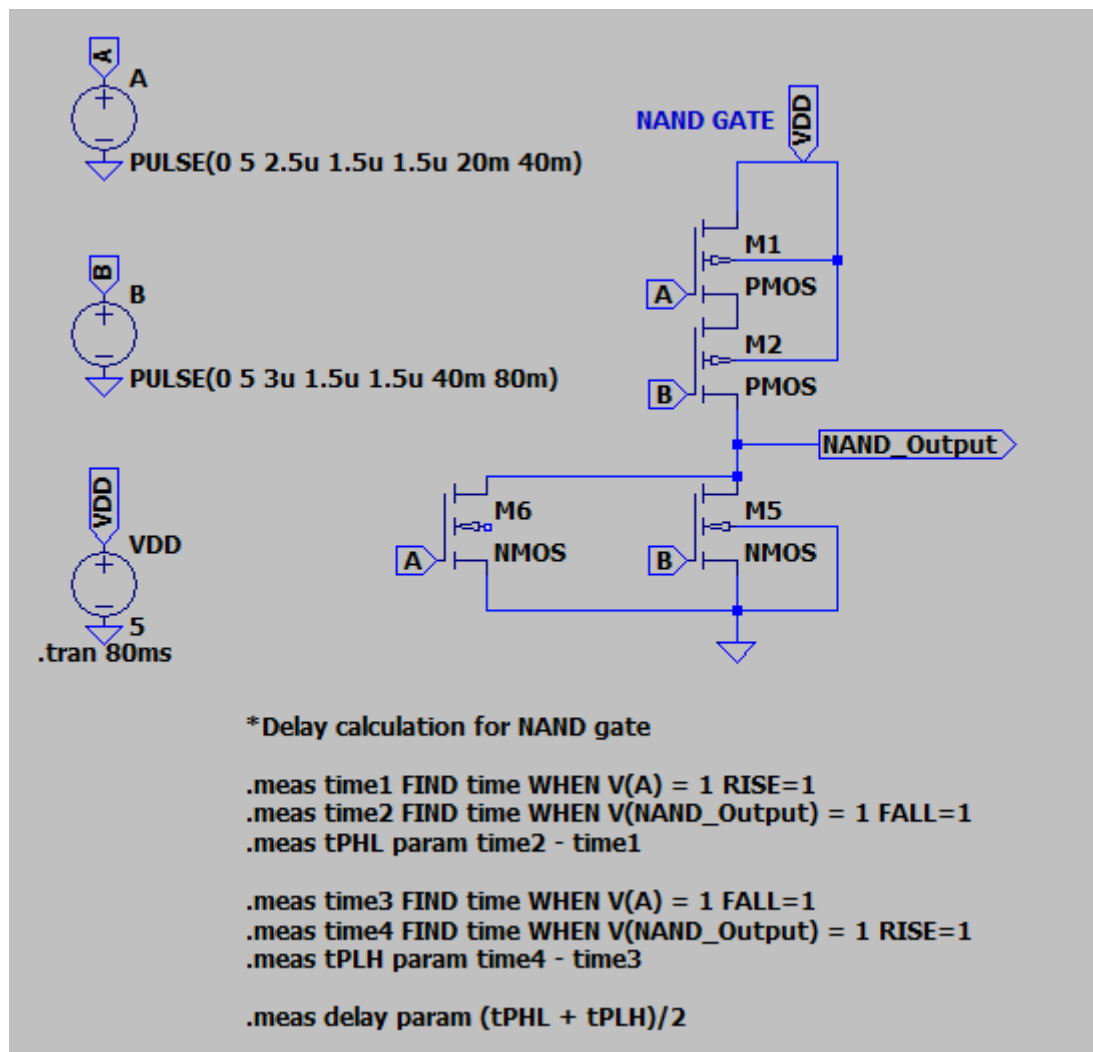
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Delay calculation:-



```
time1: time=2.80000000894e-06 at 2.80000000894e-06
time2: time=3.29746729427e-06 at 3.29746729427e-06
tphl: time2 - time1=4.97467285325e-07
time3: time=0.0200052 at 0.0200052
time4: time=0.0600045295626 at 0.0600045295626
tplh: time4 - time3=0.0399993295626
delay: (tPHL + tPLH)/2=0.0199999135149
```

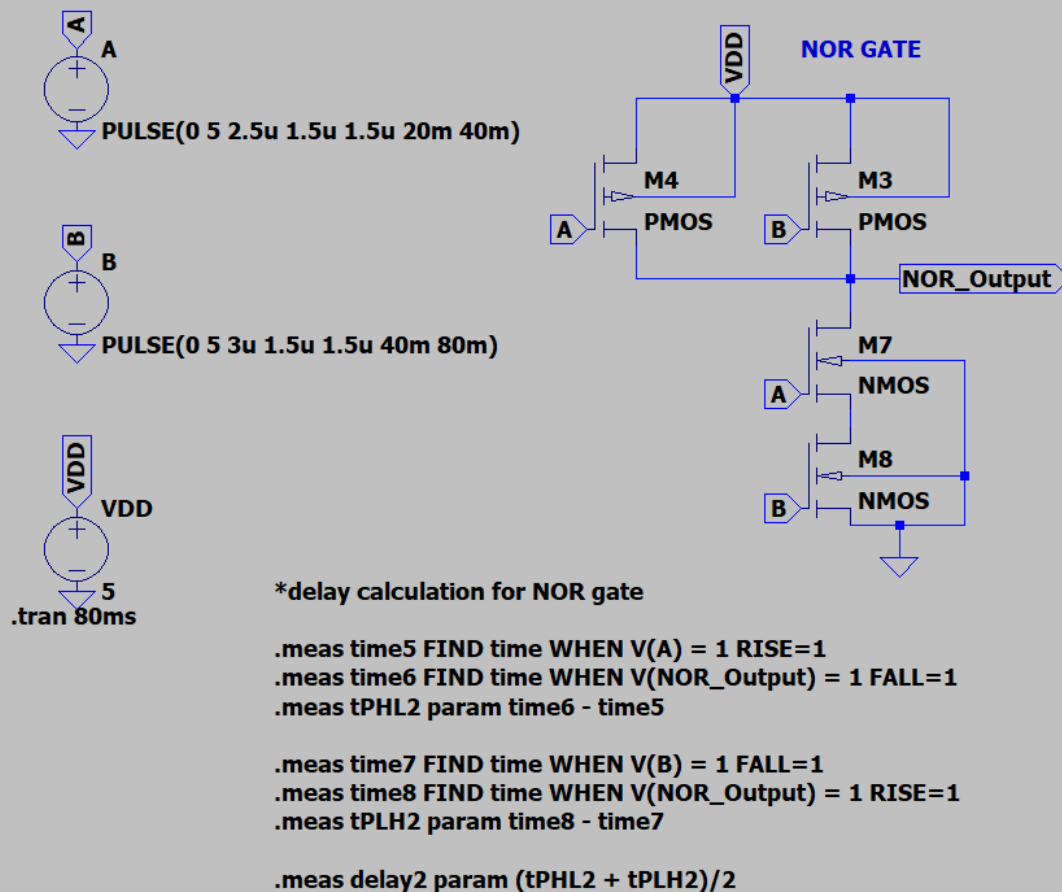
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```
time5: time=2.80000000894e-06 at 2.80000000894e-06
time6: time=3.97524507609e-06 at 3.97524507609e-06
tphl2: time6 - time5=1.17524506715e-06
time7: time=0.0400057 at 0.0400057
time8: time=0.0200044578081 at 0.0200044578081
tplh2: time8 - time7=-0.0200012421919
delay2: (tPHL2 + tPLH2)/2=-0.0100000334734
```

Conclusion:-

We designed and simulated NAND and NOR gates using MOS transistors in LTspice. By applying input pulses, we observed how the outputs switched between high and low. We measured the time it took for the output to respond, called propagation delay.